PRO MINDS

Algebra 1 — Point-Slope Form: Answer Key

Worked answers for the 10-question worksheet. Every answer was checked by script: exactly one choice in each question matches the line.

1. B — $y - 4 = (1/2)(x - 3)$. The line passes through $(-1, 2)$ and $(3, 4)$. Slope = $(4 - 2)/(3 - (-1)) = 2/4 = 1/2$. Using $(3, 4)$: $y - 4 = (1/2)(x - 3)$.

2. D — $y - 4 = -3(x - 2)$. From the table, y falls 3 for each step in x, so slope = -3. Using $(2, 4)$: $y - 4 = -3(x - 2)$.

3. A — $y - 5 = 2(x - 2)$. Slope = $(11 - 5)/(5 - 2) = 6/3 = 2$. Using $(2, 5)$: $y - 5 = 2(x - 2)$.

4. C — $y = -4x + 5$. $y + 3 = -4x + 8$, so $y = -4x + 8 - 3 = -4x + 5$.

5. B — $y = 3x + 9$. $y - 6 = 3x + 3$, so $y = 3x + 3 + 6 = 3x + 9$.

6. D — $y - 85 = 30(x - 2)$ and $y = 30x + 25$. Slope = $(175 - 85)/(5 - 2) = 90/3 = 30$ dollars per hour. Using $(2, 85)$: $y - 85 = 30(x - 2)$. Expanding: $y = 30x - 60 + 85 = 30x + 25$, so the one-time fee is $25.

7. C — $y - 3 = -(x + 2)$. Slope = $(-1 - 3)/(2 - (-2)) = -4/4 = -1$. Using $(-2, 3)$: $y - 3 = -1(x - (-2)) = -(x + 2)$.

8. A — $y - 0 = (1/2)(x - 4)$. The line passes through $(0, -2)$ and $(4, 0)$. Slope = $(0 - (-2))/(4 - 0) = 2/4 = 1/2$. Using $(4, 0)$: $y - 0 = (1/2)(x - 4)$.

9. D — $y - 4 = (3/2)(x - 1)$. From the table, y rises 1.5 for each step in x, so slope = 3/2. Using $(1, 4)$: $y - 4 = (3/2)(x - 1)$.

10. B — $y - 4 = -2(x + 3)$. Slope is given as -2 and the point is $(-3, 4)$: $y - 4 = -2(x - (-3)) = -2(x + 3)$.